

Do Order-Book States Predict Passive-Buy Toxicity? Evidence from BTC Perpetual Futures

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1 Abstract

This paper studies whether economically interpretable local order-book conditions predict short-horizon future returns and passive-buy adverse-selection risk in BTC perpetual futures. Using Binance Level 2 order-book snapshots merged with aggregate trade records, I construct a reduced-form empirical framework that summarizes local market conditions through recent directional order flow, near-touch bid-side absorption capacity, and liquidity-state fragility. The empirical design is motivated by a simple game-theoretic view of the limit order book: passive liquidity providers strategically supply or withdraw near-touch liquidity, while aggressive traders impose directional pressure that may or may not be absorbed depending on local displayed support.

The main empirical result is that a flow-adjusted bid-absorption proxy is substantially more informative than raw directional flow alone. Across pooled regressions, horizon-robustness exercises, and day-by-day subsamples, higher recent sell pressure relative to best-bid depth predicts lower short-horizon future returns and higher passive-buy adverse-selection risk. Recent order-flow imbalance also contains predictive information, but the signal is weaker and less stable than the composite pressure-versus-capacity measure. In addition, liquidity fragility matters as a local state variable, indicating that execution risk depends not only on recent aggressive flow, but also on the vulnerability of the surrounding displayed liquidity.

Taken together, the results suggest that passive-buy toxicity in crypto futures is best understood as a local imbalance between aggressive pressure and near-touch absorption capacity under fragile liquidity states. More broadly, the paper provides reduced-form evidence that short-horizon execution risk in high-frequency limit-order-book environments can be organized around economically interpretable local state variables rather than raw flow or static depth in isolation.

2 Introduction

In high-frequency limit-order-book markets, passive execution is valuable only if displayed liquidity can be supplied without being systematically picked off by informed or faster-moving order flow. This creates a central microstructure problem: when does a passive buy order become toxic? In economic terms, the issue is not simply whether aggressive selling exists, but whether recent sell-side pressure arrives in a local market environment where immediately available bid-side support is too weak to absorb it. This paper studies that problem in the context of BTC perpetual futures and asks whether short-horizon future returns and passive-buy adverse-selection risk can be explained by economically interpretable local order-book state variables.

The empirical setting is well suited to this question. Crypto perpetual futures trade in a high-frequency, continuously updated order-book environment in which displayed liquidity, aggressive trade flow, and short-horizon price responses interact at fine time scales. In such markets, passive liquidity providers face a repeated strategic problem. Aggressive traders reveal local directional pressure through executed trades, while passive traders must decide whether the currently displayed order book is deep enough and stable enough to absorb that pressure without exposing their quotes to adverse selection. From this perspective, the relevant state of the market is not captured by raw flow alone, nor by static depth alone, but by the local relationship between recent pressure, immediate absorption capacity, and surrounding liquidity fragility.

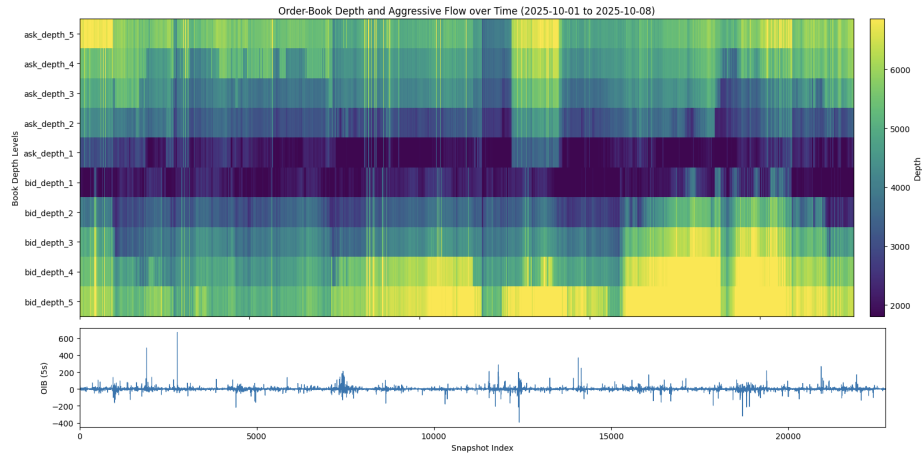


Figure 1: Order-book depth and aggressive flow over time. The upper panel plots bid-side and ask-side depth across multiple levels over the sample period, while the lower panel plots recent 5-second order-flow imbalance. The figure shows that displayed liquidity and aggressive flow vary substantially over time, suggesting that short-horizon execution risk is governed by evolving local market states rather than by static liquidity conditions.

Figure 1 provides a visual motivation for the paper. The upper panel shows that displayed liquidity at the bid and ask sides evolves continuously across depth levels, while the lower panel shows that recent aggressive order flow exhibits sharp local spikes and clustered imbalances. Taken together, these patterns suggest that the short-horizon execution environment is not static. Instead, it evolves through a sequence of local market states in which aggressive pressure and displayed liquidity interact in uneven and time-varying ways.

From a game-theoretic perspective, this is precisely the environment in which passive liquidity provision becomes a strategic problem. Aggressive traders attempt to consume liquidity when they perceive favorable local conditions, while passive liquidity providers must decide whether currently displayed depth is sufficient to absorb incoming pressure without being adversely selected. A passive quote is not exposed because the market is active in general; it is exposed because local aggressive pressure arrives when nearby displayed support is insufficient. The relevant economic object is therefore not simply whether recent order flow is positive or negative, but whether the current state of the order book can absorb that pressure.

In this sense, the market microstructure problem can be framed as a local state-mapping problem. At each snapshot time t , define a local information vector

$$X_t = (\textit{pressure}_t, \textit{capacity}_t, \textit{fragility}_t),$$

where recent order-flow imbalance proxies for pressure, best-bid depth proxies for immediate absorption capacity, and liquidity fragility proxies for the local vulnerability of displayed liquidity. The empirical objective is then to study whether

$$X_t \mapsto Y_t^{(h)}, \quad Y_t^{(h)} \in \{r_{t,t+h}, \textit{ASRisk}_t^{(h)}\},$$

for short horizons $h \in \{1s, 5s, 10s\}$.

This paper proposes a reduced-form empirical framework built around that intuition. Using Binance Level 2 order-book snapshots and aggregate trade records, I construct a set of local explanatory variables that summarize three distinct components of the market state: recent directional pressure, local pressure-versus-capacity imbalance, and background liquidity vulnerability. The first is measured by recent order-flow imbalance. The second is captured by a flow-adjusted bid-absorption proxy, defined as recent sell pressure relative to best-bid depth. The third is summarized by a liquidity-fragility measure that characterizes the local configuration of displayed depth near the touch relative to deeper levels. The central empirical question is whether these local state variables help predict short-horizon future return and passive-buy adverse-selection risk over horizons of 1 second, 5 seconds, and 10 seconds.

Formally, let

$$X_t = \left(\textit{OIB}_t^{(5s)}, \textit{Qty}_t^{(5s)}, \textit{AbsorbBid}_t^{(5s)}, \textit{Fragility}_t, \dots \right)$$

denote the vector of local market-state variables observed at snapshot time t . The econometric problem is to map X_t into short-horizon outcomes

$$Y_t^{(h)} \in \left\{ r_{t,t+h}, ASRisk_t^{(h)} \right\}, \quad h \in \{1s, 5s, 10s\},$$

where

$$r_{t,t+h} = \log M_{t+h} - \log M_t$$

is the future return and

$$ASRisk_t^{(h)} = -r_{t,t+h}$$

measures passive-buy adverse-selection risk. This setup is intentionally simple. Rather than estimating a structural continuous-time model of market making, the paper asks whether a small number of economically interpretable local variables are sufficient to explain the sign and magnitude of short-horizon outcomes.

The main empirical result is that the flow-adjusted bid-absorption proxy is substantially more informative than raw directional flow alone. Across pooled regressions, horizon robustness checks, and day-by-day subsamples, higher recent sell pressure relative to best-bid depth predicts lower future returns and higher passive-buy adverse-selection risk. By contrast, recent order-flow imbalance contains predictive information but is less stable in pooled multivariate specifications. Liquidity fragility also enters as a meaningful state variable, suggesting that local book configuration matters for execution quality even when formal interaction effects are weaker in the current reduced-form setting. Descriptive grouped-mean plots and smoothed conditional surfaces further show that the most adverse passive-buy outcomes are concentrated in regions where recent sell pressure is large relative to near-touch bid-side support and the surrounding liquidity state is fragile.

The contribution of the paper is therefore twofold. First, it provides reduced-form evidence that short-horizon passive-buy toxicity in crypto futures is better explained by local pressure-versus-capacity conditions than by raw flow or static depth in isolation. Second, it offers an economically interpretable local-state framework that connects aggressive trade flow, displayed liquidity, and execution risk in a form that is useful for high-frequency microstructure analysis. In this sense, the paper is not merely a short-horizon prediction exercise. It is an attempt to organize passive execution risk around a small set of local variables that admit a clear strategic interpretation in the limit order book.

The remainder of the paper proceeds as follows. Section 3 describes the data and variable construction. Section 4 presents the empirical design and hypotheses. Section 5 reports the main regression results and robustness checks. Section 6 provides descriptive visualizations and economic interpretation. The appendix contains supplementary derivations and additional descriptive evidence.

3 Data and Variable Construction

3.1 Data Sources and Sample Construction

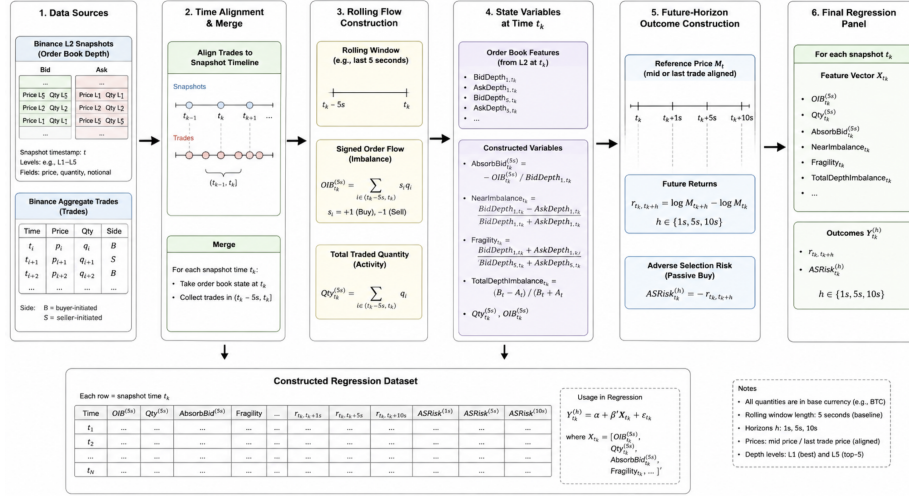


Figure 2: Data construction pipeline. The figure summarizes the construction of the regression dataset from Binance L2 order-book snapshots and aggregate trade records. Snapshot-level book states are merged with recent aggressive trade flow to construct interpretable local market variables, which are then aligned with short-horizon future return and passive-buy adverse-selection-risk outcomes.

This study uses high-frequency BTCUSDT perpetual futures data from Binance, constructed from two complementary sources: Level 2 order-book depth snapshots and aggregate trade records. The depth snapshots provide displayed bid-side and ask-side liquidity at multiple price levels, while the aggregate trades data allow the construction of recent aggressive trading flow. The empirical sample is built by merging these two sources into a common high-frequency panel indexed by snapshot timestamps.

The order-book dataset contains quantities and notionals across several bid and ask levels, which are used to characterize the local liquidity environment. The aggregate trades dataset contains trade direction and traded quantity information, which are used to construct recent order-flow pressure variables over rolling windows. After time alignment and cleaning, the merged sample provides a sequence of local market states suitable for studying short-horizon return formation and passive-buy adverse-selection risk.

3.2 Time Alignment and Snapshot-Based Sampling

Let t denote a snapshot timestamp from the L2 order-book data. For each snapshot time t , I align recent aggressive trade flow and future price outcomes relative to that timestamp. This produces a panel in which the explanatory variables summarize local order-book and trade-flow conditions at time t , while the dependent variables measure the subsequent short-horizon price movement over horizon h .

Formally, let M_t denote the reference price at time t , measured using the transaction-aligned price in the merged dataset. For each horizon $h \in \{1s, 5s, 10s\}$, I define future outcomes relative to the current snapshot time. The resulting empirical design is therefore forward-looking in the dependent variable and contemporaneous in the explanatory variables.

This snapshot-based design is useful because the dependent variables are defined over future horizons, whereas the explanatory variables summarize the local information set available at time t . In this sense, the empirical specification is intended to examine whether local microstructure conditions help predict short-horizon outcomes.

3.3 Dependent Variables

The first dependent variable is the short-horizon future return over horizon h , defined as

$$r_{t,t+h} = \log M_{t+h} - \log M_t, \quad h \in \{1s, 5s, 10s\}. \quad (1)$$

This variable measures the signed short-horizon price movement following the current snapshot. A positive value means that the price rises after time t , while a negative value means that the price falls. The second dependent variable is passive-buy adverse-selection risk, defined as the negative of the future return:

$$ASRisk_t^{(h)} = -r_{t,t+h} = -(\log M_{t+h} - \log M_t). \quad h \in \{1s, 5s, 10s\}. \quad (2)$$

This definition is motivated by the economics of passive buy execution. A passive buy order is exposed to adverse selection when it is filled and the market subsequently moves downward. Under this definition, larger values of $ASRisk_t^{(h)}$ correspond to more adverse passive-buy outcomes.

The relationship between the two dependent variables is therefore immediate:

$$ASRisk_t^{(h)} = -r_{t,t+h}. \quad (3)$$

Thus, any predictor that enters positively for future return should enter negatively for adverse-selection risk, and vice versa. This symmetry provides a useful consistency check for the empirical results.

3.4 Core Explanatory Variables

3.4.1 Order-Flow Imbalance

To capture recent directional aggressive trading pressure, I construct a recent order-flow imbalance variable over the previous 5 seconds:

$$OIB_t^{(5s)} = \sum_{i \in (t-5s, t]} s_i q_i, \quad (4)$$

where q_i denotes traded quantity for trade i , and $s_i \in \{+1, -1\}$ denotes trade direction, with positive sign for buyer-initiated trades and negative sign for seller-initiated trades.

Economically, $OIB_t^{(5s)}$ measures the net aggressive pressure over the recent past. A larger positive value indicates net buyer-initiated pressure, while a more negative value indicates net seller-initiated pressure.

3.4.2 Trading Quantity

To control for the overall level of recent market activity, I also construct total traded quantity over the previous 5 seconds:

$$Qty_t^{(5s)} = \sum_{i \in (t-5s, t]} q_i. \quad (5)$$

This variable measures the intensity of recent trading activity without imposing any directional interpretation. It is included primarily as an activity control rather than as a core mechanism variable.

3.4.3 Near-Touch Bid Depth

Let $BidDepth_{1,t}$ denote displayed bid-side depth at the best bid at time t . This is the immediate local liquidity available to absorb sell-side aggressive pressure. Because passive buy orders are executed at or near the bid, the best-bid depth is particularly relevant for studying passive-buy execution risk.

3.4.4 Flow-Adjusted Bid Absorption Pressure

The key mechanism variable in the paper is a flow-adjusted bid absorption proxy, defined as

$$AbsorbBid_t^{(5s)} = \frac{-OIB_t^{(5s)}}{BidDepth_{1,t}}. \quad (6)$$

This variable is designed to capture recent seller-initiated pressure relative to local bid-side absorption capacity. The logic is straightforward. When $OIB_t^{(5s)}$ is negative, recent aggressive selling pressure is strong. Dividing by $BidDepth_{1,t}$ scales that pressure by the amount of immediate bid-side liquidity available to absorb it.

The variable can therefore be interpreted as a local pressure-versus-capacity ratio. A large value in magnitude indicates that recent sell pressure is large relative to near-touch bid capacity. Economically, this is precisely the environment in which passive buy orders are more likely to be adversely selected.

This construction is not intended as a structural model of execution, but as an interpretable reduced-form proxy that combines two economically meaningful ingredients: recent aggressive pressure and local displayed liquidity.

3.4.5 Near-Touch Imbalance

To measure local asymmetry in displayed liquidity at the top of the book, I construct a near-touch imbalance variable:

$$NearImbalance_t = \frac{BidDepth_{1,t} - AskDepth_{1,t}}{BidDepth_{1,t} + AskDepth_{1,t}}. \quad (7)$$

This variable ranges between -1 and 1 . Positive values indicate relatively stronger bid-side depth at the best quotes, while negative values indicate relatively stronger ask-side depth.

3.4.6 Total Depth Imbalance

To summarize broader displayed asymmetry across depth, I construct a total depth imbalance variable:

$$TotalDepthImbalance_t = \frac{B_t - A_t}{B_t + A_t}, \quad (8)$$

where B_t and A_t denote total bid-side and ask-side displayed depth across the included levels at time t . This variable measures whether the book is bid-heavy or ask-heavy at a broader local scale.

3.4.7 Liquidity Fragility

To characterize the local configuration of displayed liquidity, I define near-touch total depth and deeper total depth as

$$NearTotalDepth_t = BidDepth_{1,t} + AskDepth_{1,t}, \quad (9)$$

$$FarTotalDepth_t = BidDepth_{5,t} + AskDepth_{5,t}, \quad (10)$$

and then construct liquidity fragility as

$$Fragility_t = \frac{NearTotalDepth_t}{FarTotalDepth_t}. \quad (11)$$

This variable is intended to summarize the extent to which displayed liquidity is concentrated near the touch relative to deeper levels. In the paper, it is interpreted as a local liquidity-state variable rather than a direct pressure

variable. It characterizes the background condition of the order book, while $OIB_t^{(5s)}$ and $AbsorbBid_t^{(5s)}$ characterize recent flow pressure.

3.5 Economic Interpretation of the Variables

The explanatory variables are designed to map into a simple game-theoretic interpretation of the limit order book. The local market environment at time t can be viewed as the outcome of a strategic interaction between aggressive traders and passive liquidity providers.

Recent order-flow imbalance, $OIB_t^{(5s)}$, measures who has been applying directional pressure over the recent past. However, aggressive pressure alone need not imply high execution risk. What matters for passive-buy toxicity is whether the local bid side can absorb that pressure. This is why the flow-adjusted absorption proxy,

$$AbsorbBid_t^{(5s)} = \frac{-OIB_t^{(5s)}}{BidDepth_{1,t}}, \quad (12)$$

plays a central role in the analysis. It captures the idea that passive buy orders are most vulnerable when seller-initiated pressure is strong relative to immediately available bid-side support.

Liquidity fragility, in contrast, is interpreted as a state variable describing the configuration of local displayed liquidity. It does not directly measure aggressive pressure, but rather the vulnerability of the surrounding book structure. In this sense, the empirical design separates three conceptually distinct objects: recent directional pressure, local absorption capacity, and background liquidity state.

The resulting framework is reduced-form rather than structural, but it is intentionally constructed to preserve economic interpretability. Rather than relying on black-box prediction, the empirical analysis focuses on whether these economically motivated local state variables help explain short-horizon future returns and passive-buy adverse-selection risk.

4 Empirical Design and Hypotheses

4.1 Empirical Objective

The empirical objective of this paper is to examine whether economically interpretable local order-book and trade-flow variables help predict short-horizon future returns and passive-buy adverse-selection risk in BTC perpetual futures. The analysis is intentionally reduced-form. Rather than estimating a structural model of execution, I ask whether a small set of local market-state variables observed at time t contains predictive information for subsequent outcomes over short horizons.

Formally, let

$$X_t = \left(OIB_t^{(5s)}, Qty_t^{(5s)}, AbsorbBid_t^{(5s)}, Fragility_t, NearImbalance_t, TotalDepthImbalance_t \right)$$

denote the vector of local explanatory variables at snapshot time t . The econometric question is whether these variables help explain

$$Y_t^{(h)} \in \left\{ r_{t,t+h}, ASRisk_t^{(h)} \right\}, \quad h \in \{1s, 5s, 10s\}.$$

The design emphasizes interpretability over unrestricted specification search. I first study parsimonious baseline regressions using raw directional flow, then compare them with mechanism-oriented specifications using a flow-adjusted absorption proxy, and finally augment the baseline with a liquidity-state variable.

4.2 Baseline Specification

The baseline model examines whether recent directional order flow has predictive content for short-horizon outcomes:

$$Y_t^{(h)} = \alpha + \beta_1 OIB_t^{(5s)} + \beta_2 Qty_t^{(5s)} + \varepsilon_t, \quad h \in \{1s, 5s, 10s\}. \quad (13)$$

Here, $OIB_t^{(5s)}$ captures recent signed aggressive pressure, while $Qty_t^{(5s)}$ controls for the overall intensity of recent trading activity. This specification asks whether raw directional pressure alone helps explain subsequent returns and passive-buy toxicity.

4.3 Mechanism Specification

The key mechanism model replaces raw directional flow with the flow-adjusted bid absorption proxy:

$$Y_t^{(h)} = \alpha + \beta_1 AbsorbBid_t^{(5s)} + \beta_2 Qty_t^{(5s)} + \varepsilon_t, \quad h \in \{1s, 5s, 10s\}. \quad (14)$$

This specification is motivated by the idea that passive-buy adverse selection depends not only on recent aggressive selling pressure, but on that pressure relative to the local bid-side capacity available to absorb it. If passive buy orders become most vulnerable when recent sell pressure overwhelms near-touch bid support, then $AbsorbBid_t^{(5s)}$ should be more informative than raw flow alone.

4.4 Book-State Augmented Specification

To examine whether the local liquidity environment matters beyond directional flow, I augment the baseline model with liquidity fragility and an interaction term:

$$Y_t^{(5s)} = \alpha + \beta_1 OIB_t^{(5s)} + \beta_2 Fragility_t + \beta_3 \left(OIB_t^{(5s)} \times Fragility_t \right) + \beta_4 Qty_t^{(5s)} + \varepsilon_t \quad (15)$$

This specification allows the effect of recent directional pressure to vary with the surrounding liquidity state. In economic terms, it asks whether the

same amount of recent order-flow pressure has different implications for future returns and adverse-selection risk depending on whether the order book is locally resilient or fragile.

4.5 Hypotheses

The empirical analysis is organized around four main hypotheses.

Hypothesis 1: Directional flow. Recent order-flow imbalance contains predictive information for short-horizon outcomes:

$$H_0 : \beta_{OIB} = 0 \quad \text{versus} \quad H_1 : \beta_{OIB} \neq 0.$$

For future return regressions, the expected sign is

$$\beta_{OIB} > 0,$$

while for adverse-selection-risk regressions, the expected sign is

$$\beta_{OIB} < 0.$$

Hypothesis 2: Flow-adjusted absorption pressure. Recent sell pressure relative to near-touch bid capacity is more informative than raw flow alone:

$$H_0 : \beta_{AbsorbBid} = 0 \quad \text{versus} \quad H_1 : \beta_{AbsorbBid} \neq 0.$$

For future return regressions, the expected sign is

$$\beta_{AbsorbBid} < 0,$$

while for adverse-selection-risk regressions, the expected sign is

$$\beta_{AbsorbBid} > 0.$$

Hypothesis 3: Liquidity state. Local liquidity fragility matters as a standalone state variable:

$$H_0 : \beta_{Fragility} = 0 \quad \text{versus} \quad H_1 : \beta_{Fragility} \neq 0$$

Hypothesis 4: State-dependent flow effects. The marginal effect of recent order-flow pressure may vary across liquidity states:

$$H_0 : \beta_{OIB \times Fragility} = 0 \quad \text{versus} \quad H_1 : \beta_{OIB \times Fragility} \neq 0.$$

4.6 Estimation and Inference

All regressions are estimated using ordinary least squares with heteroskedasticity-robust standard errors. Because raw price levels in high-frequency financial data may exhibit near-unit-root behavior, the main dependent variables are defined in terms of short-horizon future returns and adverse-selection-risk measures rather than price levels. Likewise, the core explanatory variables are constructed as local flow, ratio, and liquidity-state variables, which are more naturally interpreted as short-horizon state measures than as trending levels.

The empirical strategy also includes two robustness dimensions. First, the main specifications are estimated across multiple forecasting horizons, $h \in \{1s, 5s, 10s\}$, in order to examine whether predictive content is concentrated at only one horizon or remains broadly stable across short windows. Second, the mechanism specification is re-estimated separately by trading day to verify that the pooled results are not driven by a single isolated day.

5 Results

5.1 Baseline and Mechanism Regressions

Table 1 reports the core pooled regressions for 5-second outcomes. I begin with a baseline specification using recent order-flow imbalance and recent traded quantity, and then compare it with a mechanism-oriented specification based on the flow-adjusted bid absorption proxy.

The baseline results suggest that recent directional order flow contains some predictive information for short-horizon outcomes. In the return regression, the coefficient on $OIB_t^{(5s)}$ is positive, while in the adverse-selection-risk regression it is negative, consistent with the expected sign pattern. However, the statistical strength of this effect is modest in the pooled 5-second specification, and recent traded quantity is not significant.

By contrast, the mechanism specification delivers substantially stronger evidence. The coefficient on $AbsorbBid_t^{(5s)}$ is strongly significant with the expected sign in both regressions: higher recent sell pressure relative to best-bid depth predicts lower future returns and higher passive-buy adverse-selection risk. The quantity control remains statistically weak. Taken together, these results suggest that local pressure-versus-capacity conditions are more informative than raw directional flow alone.

Table 1: Baseline and mechanism regressions for 5-second outcomes

	Return (Baseline)	ASRisk (Baseline)	Return (Mechanism)	ASRisk (Mechanism)
Main variable	3.185×10^{-7}	-3.185×10^{-7}	-1.530×10^{-6}	1.530×10^{-6}
<i>p</i> -value	0.0705	0.0705	< 0.001	< 0.001
Qty	-1.645×10^{-7}	1.645×10^{-7}	-1.198×10^{-7}	1.198×10^{-7}
<i>p</i> -value	0.3416	0.3416	0.5156	0.5156
R^2	0.001732	0.001732	0.001080	0.001080
Adj. R^2	0.001644	0.001644	0.000993	0.000993
<i>N</i>	22,749	22,749	22,749	22,749

Notes: The baseline specification uses $OIB_t^{(5s)}$ and $Qty_t^{(5s)}$ as regressors. The mechanism specification replaces raw flow with $AbsorbBid_t^{(5s)}$. All reported *p*-values are based on heteroskedasticity-robust standard errors.

5.2 Raw Components versus Composite Mechanism

To examine whether the predictive power of the absorption proxy is simply inherited from its raw components, I compare a specification using recent order-flow imbalance and best-bid depth separately with an augmented specification that additionally includes $AbsorbBid_t^{(5s)}$. Table 2 reports the results.

When recent order-flow imbalance and best-bid depth enter separately, neither variable is statistically strong in the pooled sample. Once the composite absorption proxy is added, however, the coefficient on $AbsorbBid_t^{(5s)}$ remains overwhelmingly significant, while the raw components remain weak. This finding suggests that the economically relevant reduced-form object is not aggressive flow or best-bid depth in isolation, but the local ratio of pressure to immediate absorption capacity.

This comparison is important for interpretation. It implies that adverse passive-buy outcomes are not well described by directional flow alone, nor by static near-touch depth alone. Instead, the most informative predictor is the imbalance between recent aggressive sell pressure and local displayed bid-side support.

Table 2: Raw components versus composite mechanism

	Return: Components	Return: + AbsorbBid	ASRisk: Components	ASRisk: + AbsorbBid
$OIB_t^{(5s)}$ coefficient	2.646×10^{-7}	2.639×10^{-7}	-2.646×10^{-7}	-2.639×10^{-7}
<i>p</i> -value	0.1911	0.1923	0.1911	0.1923
$BidDepth_{1,t}$ coefficient	-1.895×10^{-9}	-1.725×10^{-9}	1.895×10^{-9}	1.725×10^{-9}
<i>p</i> -value	0.2938	0.3374	0.2938	0.3374
$AbsorbBid_t^{(5s)}$ coefficient	—	-1.512×10^{-6}	—	1.512×10^{-6}
<i>p</i> -value	—	< 0.001	—	< 0.001
R^2	0.001007	0.001636	0.001007	0.001636
Adj. R^2	0.000919	0.001504	0.000919	0.001504
<i>N</i>	22,749	22,749	22,749	22,749

Notes: “Components” denotes the specification with $OIB_t^{(5s)}$ and $BidDepth_{1,t}$. “+ AbsorbBid” adds the composite pressure versus capacity proxy $AbsorbBid_t^{(5s)}$. All reported *p*-values are based on heteroskedasticity-robust standard errors.

5.3 Horizon Robustness

I next evaluate whether the reduced-form results remain stable across forecasting horizons. The same 5-second explanatory variables are used to forecast 1-second, 5-second, and 10-second outcomes. Table 3 summarizes the results.

The baseline flow model shows that recent order-flow imbalance enters with the expected sign across all three horizons. The effect is strongest at the 1-second and 10-second horizons and becomes weaker in the pooled 5-second specification. By contrast, the mechanism model based on $AbsorbBid_t^{(5s)}$ is directionally stable and highly significant across all horizons for both future returns and adverse-selection risk.

This pattern supports a central theme of the paper: raw directional flow contains predictive information, but flow-adjusted local absorption pressure is the more stable and economically interpretable reduced-form predictor of passive-buy toxicity. Recent traded quantity remains weak across nearly all specifications.

Table 3: Horizon robustness of baseline and mechanism models

Horizon	Dependent variable	Main variable	Main coefficient	Main p -value	R^2
1s	Return	$OIB_t^{(5s)}$	2.251×10^{-7}	0.0003	0.004471
1s	ASRisk	$OIB_t^{(5s)}$	-2.251×10^{-7}	0.0003	0.004471
1s	Return	$AbsorbBid_t^{(5s)}$	-9.755×10^{-7}	< 0.001	0.001913
1s	ASRisk	$AbsorbBid_t^{(5s)}$	9.755×10^{-7}	< 0.001	0.001913
5s	Return	$OIB_t^{(5s)}$	3.185×10^{-7}	0.0705	0.001732
5s	ASRisk	$OIB_t^{(5s)}$	-3.185×10^{-7}	0.0705	0.001732
5s	Return	$AbsorbBid_t^{(5s)}$	-1.530×10^{-6}	< 0.001	0.001080
5s	ASRisk	$AbsorbBid_t^{(5s)}$	1.530×10^{-6}	< 0.001	0.001080
10s	Return	$OIB_t^{(5s)}$	6.050×10^{-7}	0.0023	0.003045
10s	ASRisk	$OIB_t^{(5s)}$	-6.050×10^{-7}	0.0023	0.003045
10s	Return	$AbsorbBid_t^{(5s)}$	-1.985×10^{-6}	< 0.001	0.001328
10s	ASRisk	$AbsorbBid_t^{(5s)}$	1.985×10^{-6}	< 0.001	0.001328

Notes: The baseline model uses $OIB_t^{(5s)}$ and $Qty_t^{(5s)}$ as regressors. The mechanism model uses $AbsorbBid_t^{(5s)}$ and $Qty_t^{(5s)}$. Reported coefficients and p -values correspond to the main explanatory variable in each specification.

5.4 By-Day Robustness

A potential concern is that the pooled mechanism results are driven by one or two unusually strong sample days. To address this issue, I re-estimate the mechanism specification separately by trading day for each horizon and each dependent variable. Table 4 reports summary statistics for the day-by-day coefficients.

The coefficient on $AbsorbBid_t^{(5s)}$ is directionally stable across most days.

For future returns, the coefficient is negative on the majority of days, and for adverse-selection risk it is positive on the majority of days. This sign pattern is strongest at the 5-second and 10-second horizons, where the expected-sign share reaches 87.5%. The share of individually significant daily regressions is lower than in the pooled sample but still economically significant given the smaller per-day sample sizes.

Overall, the day-by-day evidence supports the interpretation that the pooled mechanism result is not driven by a single isolated episode. Rather, the pressure-versus-capacity effect appears repeatedly across days, even though its day-level statistical strength varies.

Table 4: By-day robustness summary for the mechanism model

Outcome	Mean coef.	Median coef.	Exp. sign share	$p < 0.10$	$p < 0.05$
Ret (1s)	-0.000361	-0.000458	0.750	0.625	0.625
ASR (1s)	0.000361	0.000458	0.750	0.625	0.625
Ret (5s)	-0.000669	-0.000694	0.875	0.500	0.375
ASR (5s)	0.000669	0.000694	0.875	0.500	0.375
Ret (10s)	-0.001292	-0.001434	0.875	0.625	0.500
ASR (10s)	0.001292	0.001434	0.875	0.625	0.500

Notes: “Ret” denotes future return and “ASR” denotes adverse-selection risk. The table summarizes day-by-day regressions of $Y_t^{(h)}$ on $AbsorbBid_t^{(5s)}$ and $Qty_t^{(5s)}$ across eight sample days. “Expected sign” means negative for return regressions and positive for adverse-selection-risk regressions.

6 Descriptive Visualizations and Economic Interpretation

This section complements the regression analysis with descriptive figures that illustrate the distributional properties of the core explanatory variables and the conditional patterns underlying the main empirical results. The purpose of these figures is not to provide formal inference, but to clarify the economic interpretation of the reduced-form estimates. In particular, the visualizations help show how short-horizon future returns and passive-buy adverse-selection risk vary between local pressure, absorption-capacity, and liquidity-state conditions.

6.1 Distributional Properties of the Core Variables

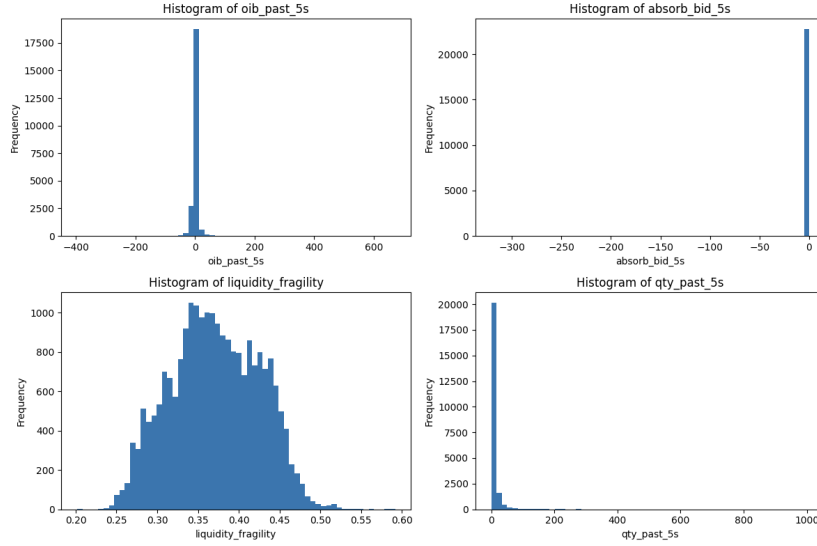


Figure 3: Unconditional distributions of the core explanatory variables.

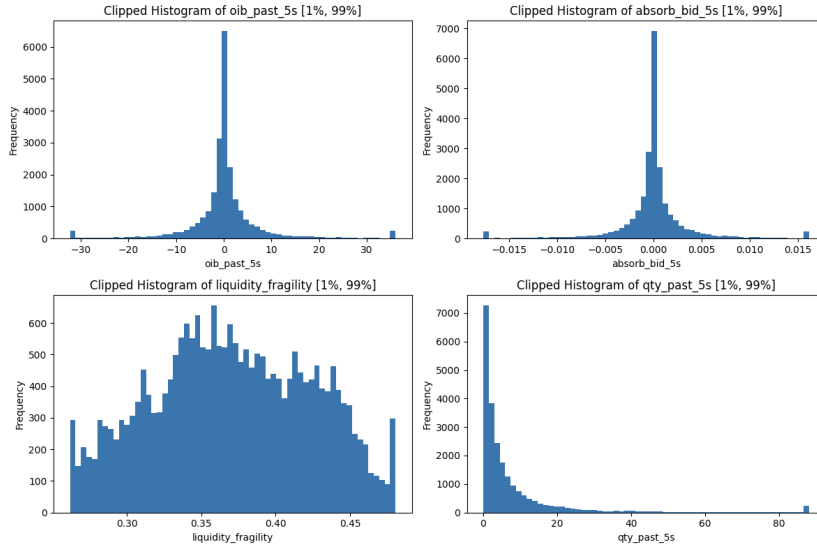


Figure 4: Clipped distributions of the core explanatory variables.

Figure 3 reports the unconditional distributions of the core explanatory variables, while Figure 4 reports the corresponding clipped distributions after trim-

ming the most extreme tails for visualization. Two features stand out. First, both recent order-flow imbalance and the flow-adjusted bid-absorption proxy are highly concentrated near zero, with relatively infrequent but economically meaningful tail realizations. This indicates that the local market is often near short-run directional balance, but occasionally experiences episodes of unusually strong aggressive pressure. Second, traded quantity is strongly right-skewed, suggesting that most 5-second windows contain modest activity while a small number of windows are characterized by unusually intense trading. By contrast, liquidity fragility is more tightly distributed and behaves less like a shock variable and more like a local state variable.

These distribution patterns are economically important. In a simple game-theoretic interpretation of the limit order book, aggressive traders do not arrive at a constant intensity, and passive liquidity providers do not face constant execution risk. Instead, short-horizon toxicity is concentrated in relatively rare local states in which the recent pressure becomes unusually large relative to the nearby displayed capacity. The shape of the distributions is therefore consistent with the interpretation that adverse selection is episodic rather than uniform.

6.2 Grouped Means and One-Dimensional Economic Gradients

Figure 5 plots grouped means of future return and adverse-selection risk against the flow-adjusted bid-absorption proxy. The pattern is economically intuitive and closely aligned with the regression results. As the local pressure-versus-capacity measure becomes more adverse, mean future returns deteriorate and mean passive-buy adverse-selection risk rises. This suggests that passive buy orders become more vulnerable precisely when recent sell-side aggressive pressure is large relative to the best-bid depth available to absorb it.

Figure 6 shows the corresponding grouped means for recent order-flow imbalance. The relationship is directionally sensible: more positive recent order-flow imbalance is associated with higher future returns and lower passive-buy adverse-selection risk. However, the pattern is less sharp than for the absorption proxy. This is consistent with the regression evidence that raw directional flow contains predictive information, but that flow-adjusted local absorption conditions are more informative than directional flow alone.

Taken together, these grouped-mean figures show that the main empirical relations are visible even before imposing a fully parametric specification. In this sense, the visual evidence supports the reduced-form interpretation advanced in Section 5 rather than replacing it.

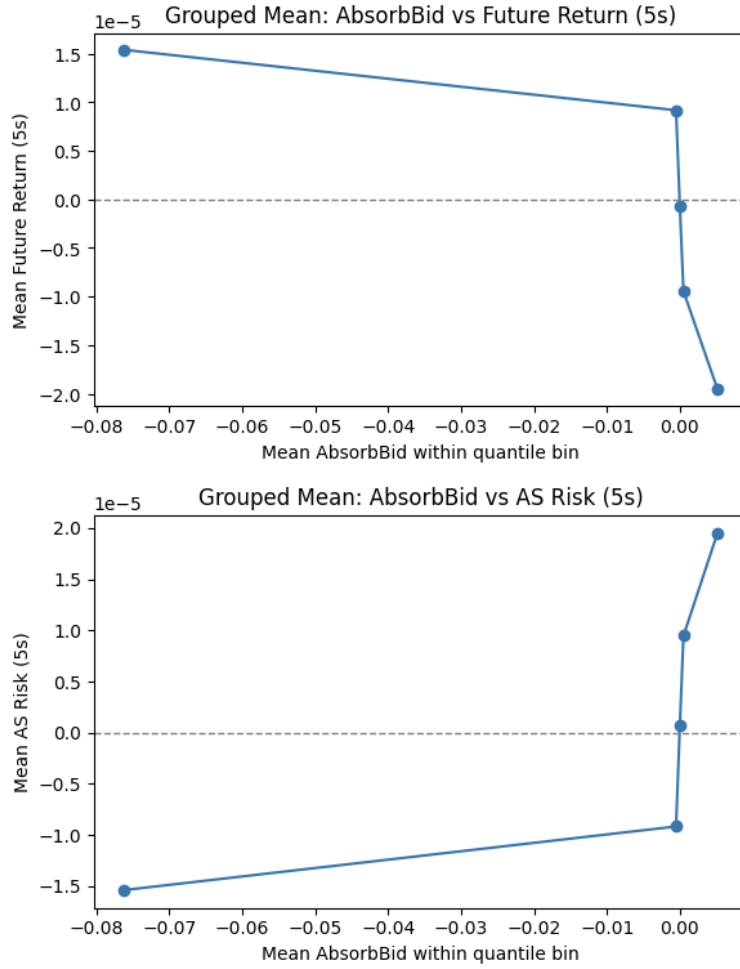


Figure 5: Grouped means of short-horizon outcomes against the flow-adjusted bid-absorption proxy. The upper panel plots mean 5-second future return within quantile bins of $AbsorbBid_t^{(5s)}$, while the lower panel plots mean 5-second passive-buy adverse-selection risk. The figure shows that more adverse local pressure-versus-capacity conditions are associated with lower future returns and higher passive-buy toxicity.

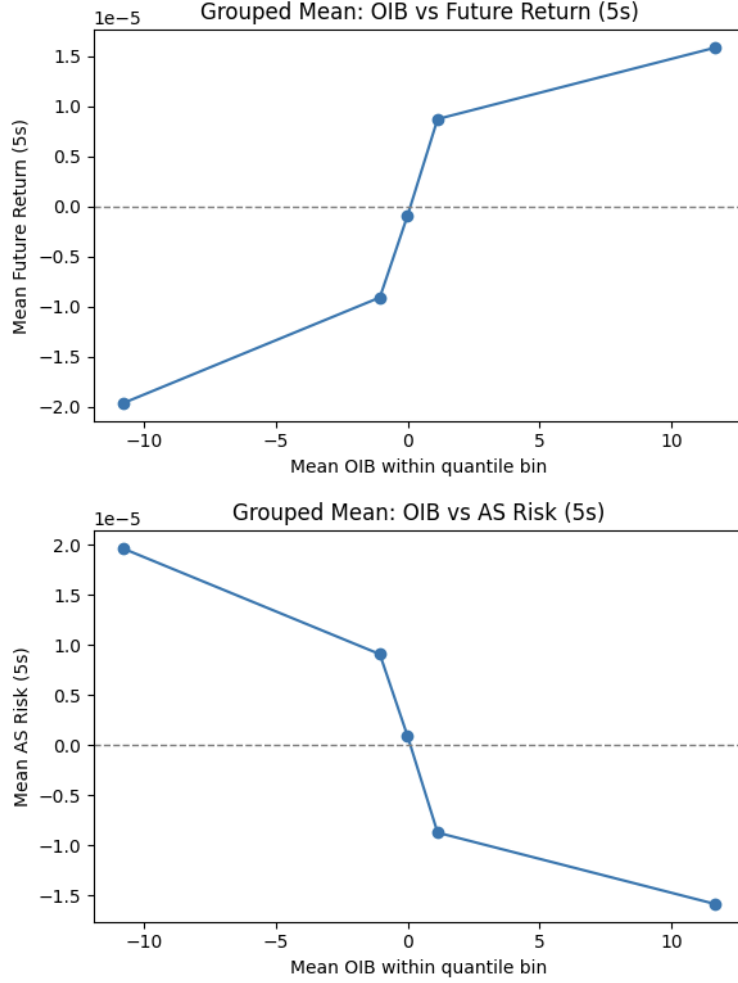


Figure 6: Grouped means of short-horizon outcomes against recent order-flow imbalance. The upper panel plots mean 5-second future return within quantile bins of $OIB_t^{(5s)}$, while the lower panel plots mean 5-second passive-buy adverse-selection risk. The figure indicates that raw directional flow contains predictive information, but the gradient is visually weaker than for the flow-adjusted bid-absorption proxy.

6.3 Smoothed Conditional Surfaces and State-Dependent Patterns

Figures 7 and 8 provide smoothed conditional-mean surfaces for adverse-selection risk and future return as joint functions of the flow-adjusted bid-absorption proxy and liquidity fragility. These figures offer the clearest visual summary of

the paper’s economic mechanism. The most adverse regions of the surface tend to arise when recent sell pressure relative to near-touch bid capacity is elevated and the surrounding liquidity environment is weak. In these regions, mean future returns become more negative and mean passive-buy adverse-selection risk rises.

This pattern is consistent with a simple strategic interpretation of the order book. Passive liquidity providers are not exposed merely because sellers are active. They are most exposed when sellers are active at a time when immediately available bid-side support is insufficient and the surrounding displayed liquidity is fragile. In this sense, the relevant reduced-form object is not raw flow or static depth considered separately, but the local mismatch between aggressive pressure and absorption capacity under a given liquidity state.

The smoothed surfaces are descriptive rather than inferential, and they should be interpreted accordingly. Their value lies in showing that the same economic logic appears visually across a broad range of local market states, rather than only through a single regression coefficient.

Smoothed Conditional Surface: $AbsorbBid \times Fragility \rightarrow AS\ Risk\ (5s)$

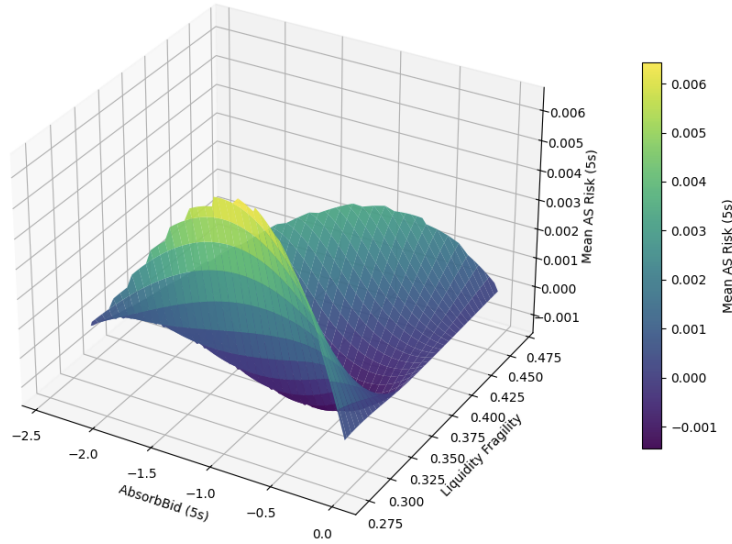


Figure 7: Smoothed conditional surface: $AbsorbBid_t^{(5s)} \times Fragility_t \rightarrow ASRisk_t^{(5s)}$. The figure plots the conditional mean of 5-second passive-buy adverse-selection risk as a joint function of flow-adjusted sell pressure and liquidity fragility. Higher regions of the surface correspond to more adverse passive-buy outcomes and are concentrated in areas where local sell pressure is high relative to best-bid support and the surrounding liquidity state is fragile.

Smoothed Conditional Surface: AbsorbBid $\times$ Fragility $\rightarrow$ Future Return (5s)

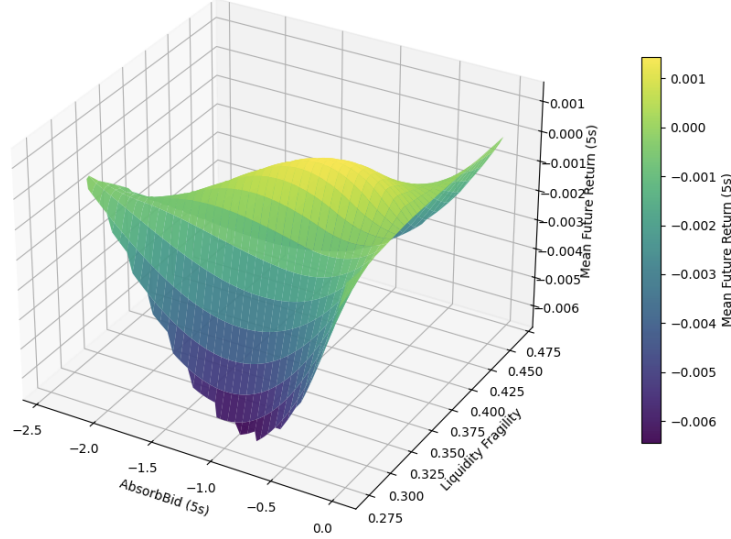


Figure 8: Smoothed conditional surface: $AbsorbBid_t^{(5s)} \times Fragility_t \rightarrow r_{t,t+5s}$. The figure plots the conditional mean of 5-second future return as a joint function of flow-adjusted sell pressure and liquidity fragility. Lower regions of the surface correspond to more negative short-horizon returns and are concentrated where recent sell pressure overwhelms local bid-side support under weaker liquidity conditions.

6.4 Why the Reduced-Form Model Explains the Visual Patterns

The visual evidence is informative precisely because it mirrors the logic of the reduced-form model. The empirical framework separates the local market environment into three conceptually distinct components: recent directional pressure, local absorption capacity, and background liquidity state. Recent directional pressure is captured by $OIB_t^{(5s)}$, local absorption capacity enters through $BidDepth_{1,t}$, and their economically relevant imbalance is summarized by $AbsorbBid_t^{(5s)}$. Liquidity fragility enters separately as a state variable describing the vulnerability of displayed liquidity.

Under this framework, the figures should not be interpreted as independent proof of the mechanism, but rather as descriptive support for it. The grouped means and conditional surfaces show that the empirical relation is not an artifact of a single regression coefficient. Instead, the same basic logic appears repeatedly across one-dimensional gradients, two-dimensional state surfaces, and pooled regression estimates: passive-buy outcomes are most adverse when aggressive

sell pressure is high relative to nearby bid-side support, especially in fragile liquidity states.

More broadly, the figures support the paper’s interpretation that passive-buy toxicity is governed by local pressure-versus-capacity conditions rather than by raw flow or static displayed depth in isolation. In this sense, the reduced-form model remains economically interpretable while still capturing the main patterns observed in the high-frequency data.

7 Appendix

7.1 Additional Descriptive Figures

Figures 9–10 provide additional descriptive visualizations for the OIB–fragility conditional surfaces. These figures are broadly consistent with the main results, but are omitted from the main text in order to keep the central narrative focused on the flow-adjusted absorption mechanism.

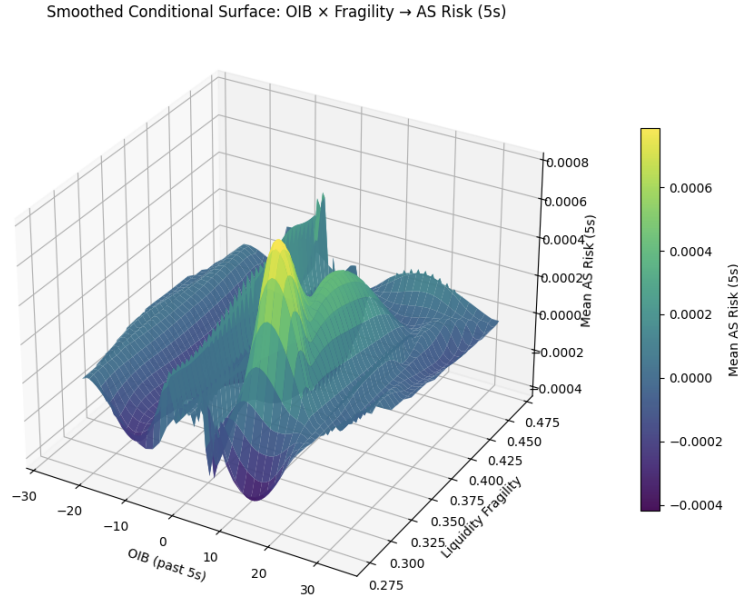


Figure 9: Smoothed conditional surface: $OIB_t^{(5s)} \times Fragility_t \rightarrow ASRisk_t^{(5s)}$. The figure plots the conditional mean of 5-second passive-buy adverse-selection risk as a joint function of recent order-flow imbalance and liquidity fragility. The surface suggests state-dependent variation in passive-buy toxicity, but the structure is visually less sharp than in the absorption-based specification shown in the main text.

Smoothed Conditional Surface: $OIB \times \text{Fragility} \rightarrow \text{Future Return (5s)}$

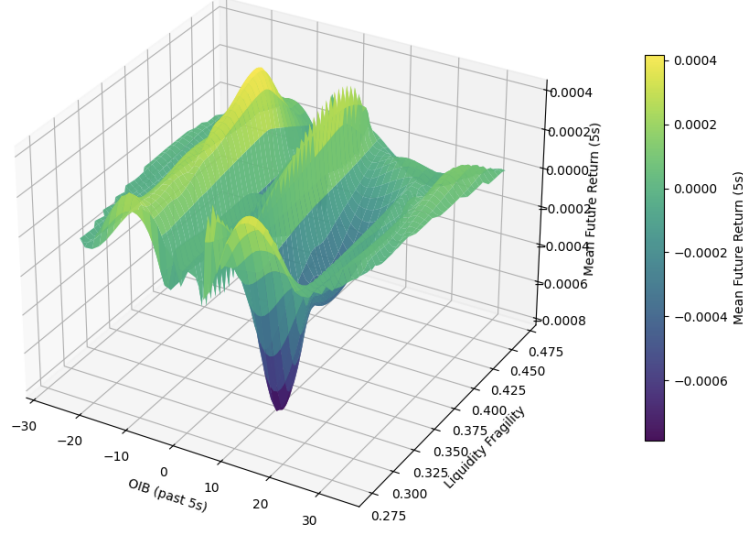


Figure 10: Smoothed conditional surface: $OIB_t^{(5s)} \times \text{Fragility}_t \rightarrow r_{t,t+5s}$. The figure plots the conditional mean of 5-second future return as a joint function of recent order-flow imbalance and liquidity fragility. The figure indicates some state-dependent variation, but the visual gradient is weaker than in the corresponding absorption-based surface.

7.2 Derivations and Variable Logic

This appendix provides brief derivations and formal interpretations of the key reduced-form variables used in the paper. The purpose is not to introduce a fully structural microstructure model, but to clarify why the main explanatory variables can be interpreted as economically meaningful local state measures.

7.3 Why Adverse-Selection Risk Is Defined as the Negative of Future Return

Let the short-horizon future return over horizon h be defined as

$$r_{t,t+h} = \log M_{t+h} - \log M_t. \quad (16)$$

For a passive buy order, adverse selection occurs when the order is executed and the subsequent market price moves downward. In that case, a more negative future return corresponds to a more adverse execution outcome. This motivates the passive-buy adverse-selection measure

$$ASRisk_t^{(h)} = -r_{t,t+h}. \quad (17)$$

Hence,

$$ASRisk_t^{(h)} = -(\log M_{t+h} - \log M_t). \quad (18)$$

Under this definition, predictors that raise future return should lower adverse-selection risk, while predictors that lower future return should raise adverse-selection risk. This sign symmetry is used throughout the paper as an internal consistency check.

7.4 Why the Absorption Proxy Represents Pressure versus Capacity

Recent order-flow imbalance is defined as

$$OIB_t^{(5s)} = \sum_{i \in (t-5s, t]} s_i q_i, \quad (19)$$

where $s_i = +1$ for buyer-initiated trades and $s_i = -1$ for seller-initiated trades.

When $OIB_t^{(5s)} < 0$, recent aggressive sell pressure dominates. Let $BidDepth_{1,t}$ denote the displayed best-bid depth at time t , which represents the immediate local capacity available to absorb sell-side aggression. The flow-adjusted bid-absorption proxy is then defined as

$$AbsorbBid_t^{(5s)} = \frac{-OIB_t^{(5s)}}{BidDepth_{1,t}}. \quad (20)$$

This ratio has a natural reduced-form interpretation. The numerator measures recent sell-side pressure, while the denominator measures immediately available bid-side support. Therefore, the ratio summarizes the local mismatch between aggressive pressure and nearby absorption capacity. Larger values indicate that recent sell pressure is large relative to the liquidity available at the best bid, which is precisely the environment in which passive buy orders should face elevated adverse-selection risk.

7.5 Local State Decomposition

The reduced-form empirical design can be understood as decomposing the local market state into three distinct components:

$$\begin{aligned} \text{Directional pressure} &\longrightarrow OIB_t^{(5s)}, \\ \text{Pressure} - \text{capacity mismatch} &\longrightarrow AbsorbBid_t^{(5s)}, \\ \text{Background liquidity state} &\longrightarrow Fragility_t. \end{aligned}$$

Liquidity fragility is defined as

$$Fragility_t = \frac{NearTotalDepth_t}{FarTotalDepth_t}, \quad (21)$$

where

$$NearTotalDepth_t = BidDepth_{1,t} + AskDepth_{1,t}, \quad FarTotalDepth_t = BidDepth_{5,t} + AskDepth_{5,t}. \quad (22)$$

This construction is intended to separate transient aggressive pressure from the broader local shape of displayed liquidity. In this sense, the explanatory variables used in the paper form a reduced-form local-state system rather than a collection of unrelated signals.

7.6 Additional Binned-Scatter Evidence

This appendix reports additional binned-scatter figures for recent order-flow imbalance and short-horizon outcomes. These figures provide a nonparametric visual complement to the pooled regressions and grouped-mean evidence reported in the main text. Their role is descriptive rather than inferential, but they help verify that the directional gradients documented in the regressions are also visible in bin-level conditional means.

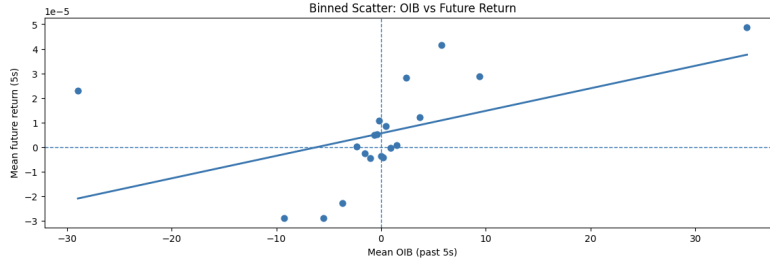


Figure 11: Binned scatter: $OIB_t^{(5s)}$ and $r_{t,t+5s}$. The figure plots mean 5-second future return against binned values of recent order-flow imbalance. The fitted line is upward sloping, consistent with the regression evidence that more positive recent order-flow imbalance predicts higher short-horizon returns.

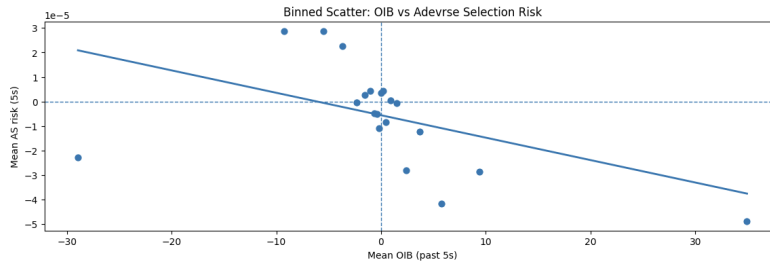


Figure 12: Binned scatter: $OIB_t^{(5s)}$ and $ASRisk_t^{(5s)}$. The figure plots mean 5-second passive-buy adverse-selection risk against binned values of recent order-flow imbalance. The fitted line is downward sloping, consistent with the sign symmetry implied by the definition $ASRisk_t^{(h)} = -r_{t,t+h}$.

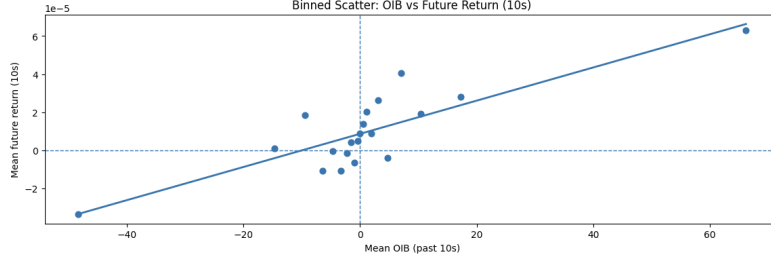


Figure 13: Binned scatter: $OIB_t^{(5s)}$ and $r_{t,t+10s}$. The figure plots mean 10-second future return against binned values of recent order-flow imbalance. The positive gradient is visually consistent with the horizon-robustness regressions reported in the main text.

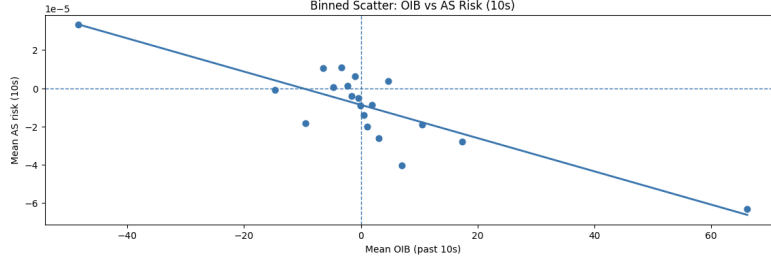


Figure 14: Binned scatter: $OIB_t^{(5s)}$ and $ASRisk_t^{(10s)}$. The figure plots mean 10-second passive-buy adverse-selection risk against binned values of recent order-flow imbalance. The negative slope provides a visual counterpart to the baseline regression evidence at the 10-second horizon.

8 Conclusion

This paper studies whether economically interpretable local order-book and trade-flow variables can explain short-horizon future returns and passive-buy adverse-selection risk in BTC perpetual futures. Using Binance Level 2 order-book snapshots merged with aggregate trade records, I construct a reduced-form empirical framework that separates local market conditions into three components: recent directional pressure, local absorption capacity, and background liquidity state. The central objective is to understand whether passive-buy toxicity is better explained by raw order flow alone or by local pressure-versus-capacity conditions embedded in the order book.

The empirical results support three main conclusions. First, recent order-flow imbalance contains short-horizon predictive information, but its explanatory strength is limited in pooled multivariate regressions. Second, a flow-adjusted bid-absorption proxy emerges as the most robust reduced-form pre-

dicator of passive-buy outcomes: higher recent sell pressure relative to best-bid depth predicts lower short-horizon future returns and higher passive-buy adverse-selection risk. Third, liquidity fragility matters as a standalone state variable, indicating that local book configuration affects execution quality even when formal interaction effects are weaker in the current specification. These patterns remain broadly stable across 1-second, 5-second, and 10-second horizons and are repeated across daily subsamples.

Taken together, the findings support a simple game-theoretic interpretation of the limit order book. Passive liquidity providers are not exposed merely because aggressive traders are active; they are most exposed when aggressive sell pressure arrives at a time when immediately available bid-side support is insufficient and the surrounding displayed liquidity is fragile. In this sense, adverse selection is shaped by a local strategic imbalance between pressure and absorption capacity rather than by directional flow or static depth viewed in isolation. The contribution of the paper is therefore not only predictive, but also interpretive: it shows that short-horizon passive-buy toxicity in crypto futures can be organized around an economically meaningful local state framework. From an execution perspective, the results suggest that passive order placement should not be evaluated solely on the basis of recent directional flow or displayed depth taken separately. What matters more is whether recent aggressive pressure is large relative to immediately available absorption capacity under the prevailing liquidity state. This interpretation is relevant for high-frequency execution, passive quote placement, and short-horizon toxicity monitoring in crypto limit order books.

A natural extension of the current reduced-form analysis is to discretize the core flow-pressure and liquidity-state variables into a finite state space and study their dynamic evolution in a Markovian framework. Such an extension would make it possible to analyze state persistence, transition probabilities, and toxic-regime duration, thereby connecting the current static reduced-form evidence to a dynamic model of local market-state transitions. More broadly, future work may extend the sample to longer horizons, additional trading regimes, or other crypto contracts in order to evaluate the external stability of the pressure-versus-capacity mechanism documented in this paper.

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